

chap_strate_STSI_ALLOC

2025-07-10

Planstratifiéaléatoiresimplesansremise(STSI)

```
require(sampling)

## Loading required package: sampling

data(rec99)
(Nh <- table(rec99$STRATLOG))

##
##   1   2   3   4
## 221 169 110  54
#1  2 3 4
# 221 169 110 54

attach(rec99)
(sdlogvac <- tapply(LOGVAC,STRATLOG,sd))

##           1           2           3           4
## 3.326514  6.865199 21.441989 64.684940
#1  2 3 4
#3.326514 6.865199 21.441989 64.684940
(sdlog <- tapply(LOG,STRATLOG,sd))

##           1           2           3           4
## 23.28442  52.79145 186.92935 787.90213
#1  2 3 4
#23.28442 52.79145 186.92935 787.90213

rec99.order <- rec99[order(STRATLOG),]

set.seed(123456)
nh <- c(10,15,20,35)
stsi.rec <- sampling::strata(rec99.order,"STRATLOG",
  size=nh,method="srswor")
stsi.data <- getdata(rec99.order, stsi.rec)

library(survey)

## Loading required package: grid
## Loading required package: Matrix
## Loading required package: survival
##
## Attaching package: 'survival'
```

```
## The following objects are masked from 'package:sampling':
##
##   cluster, strata
##
## Attaching package: 'survey'
##
## The following object is masked from 'package:graphics':
##
##   dotchart
ech.stsi<- svydesign(id=~CODE_N,strata=~STRATLOG,
  weights=rep(Nh/nh,nh),fpc=rep(nh/Nh,nh), data=stsi.data)
svytotal(~LOGVAC,ech.stsi)

##          total      SE
## LOGVAC 10231 675.51

#total SE
#LOGVAC 10955 704.92
```

ALLOCATIONS

```
data(rec99)
N <- nrow(rec99)
n <- 70

arrondi <- function(a)
+
{o <- order(a-floor(a),decreasing=T) #ordonne selon parties décimales
n <- ceiling(sum(a)) #calcule la taille totale de l'échantillon
#avec prise en compte des arrondis éventuels au niveau des nh
nbar = n-sum(floor(a)) #calcule le nombre de nh
#à arrondir à l'unité par excès
nouv <- floor(a) # arrondit tous les nh à l'unité par défaut
if (nbar!=0){nouv[o[1:nbar]] <- nouv[o[1:nbar]] + 1}
#arrondit certains nh par excès si nécessaire
#(les nh avec les plus faibles écarts à l'entier supérieur)
if (any(nouv == 0))
{print("Attention, une strate est à tirage nul")}
nouv}
```

Allocation proportionnelle

```
(Nh <- table(rec99$STRATLOG))

##
##   1   2   3   4
## 221 169 110  54

# 1 3 4
# 221 169 110 54
(n.prop <- arrondi(n*Nh/N))

##
##   1   2   3   4
## 28 21 14  7
```

```
# 1 2 3 4
# 28 21 14 7
```

Allocation optimale

```
(n.opt <- arrondi(n*Nh*sdlogvac/sum(Nh*sdlogvac)))
```

```
##
## 1 2 3 4
## 7 10 21 32
```

```
# 1 2 3 4
# 7 10 21 32
```

Allocation x-optimale

```
(n.optlog <- arrondi(n*Nh*sdlog/sum(Nh*sdlog)))
```

```
##
## 1 2 3 4
## 5 8 19 38
```

```
# 1 2 3 4
# 5 8 19 38
```

Allocation proportionnelle au total txU d'une variable auxiliaire x

```
(txh <- tapply(LOG, STRATLOG, FUN=sum))
```

```
##      1      2      3      4
## 12622 28962 58578 97152
```

```
# 1 2 3 4
# 12622 28962 58578 97152
tx <- sum(LOG)
(n.propX <- arrondi(n*txh/tx))
```

```
## 1 2 3 4
## 5 10 21 34
```

```
# 1 2 3 4
# 5 10 21 34
```

Tirage de l'échantillon et estimation

```
set.seed(123456)
stsi.prop <- sampling::strata(rec99.order, "STRATLOG", size=n.prop,
  method="srswor")
stsi.data.prop <- getdata(rec99.order, stsi.prop)
ech.stsi.prop <- svydesign(id=~CODE_N, strata=~STRATLOG,
  weights=rep(Nh/n.prop, n.prop), fpc=rep(n.prop/Nh, n.prop),
  data=stsi.data.prop)
svytotal(~LOGVAC, ech.stsi.prop)
```

```
##      total      SE
```

```
## LOGVAC 12194 2293.8
```

```
#total SE
```

```
#LOGVAC 11021 1200.3
```

Comparaison des stratégies (STSI, HT) et (SI, HT) par étude de Monte Carlo Stratification sur la variable LOG

```
set.seed(123456)
n.sim <- 10000
est.prop <- numeric(n.sim)
for (i in 1:n.sim)
{
  stsi.prop <- sampling::strata(rec99.order, "STRATLOG",
    size=n.prop, method="srswor")
  stsi.data.prop <- getdata(rec99.order, stsi.prop)
  ech.stsi.prop <- svydesign(id=~CODE_N, strata=~STRATLOG,
    weights=rep(Nh/n.prop, n.prop),
    fpc=rep(n.prop/Nh, n.prop),
    data=stsi.data.prop)
  est.prop[i] <- svytotal(~LOGVAC, ech.stsi.prop)[1]
}
```

```
mean(est.prop)
```

```
## [1] 10762.83
```

```
# [1] 10783.27
```

```
(cv.prop <- sd(est.prop)/mean(est.prop))
```

```
## [1] 0.1300911
```

```
# [1] 0.1279419
```

```
set.seed(123456)
n.sim <- 10000
est.opt <- numeric(n.sim)
for (i in 1:n.sim)
{
  stsi.opt <- sampling::strata(rec99.order, "STRATLOG",
    size=n.opt, method="srswor")
  stsi.data.opt <- getdata(rec99.order, stsi.opt)
  ech.stsi.opt <- svydesign(id=~CODE_N, strata=~STRATLOG,
    weights=rep(Nh/n.opt, n.opt),
    fpc=rep(n.opt/Nh, n.opt),
    data=stsi.data.opt)
  est.opt[i] <- svytotal(~LOGVAC, ech.stsi.opt)[1]
}
```

```
mean(est.opt)
```

```
## [1] 10756.27
```

```
(cv.opt <- sd(est.opt)/mean(est.opt))
```

```
## [1] 0.07044229
```

```

set.seed(123456)
n.sim <- 10000
est.optlog <- numeric(n.sim)
for (i in 1:n.sim)
{
  stsi.optlog<- sampling::strata(rec99.order,"STRATLOG",
    size=n.optlog,method="srswor")
  stsi.data.optlog <- getdata(rec99.order,stsi.optlog)
  ech.stsi.optlog <- svydesign(id=~CODE_N,strata=~STRATLOG,
    weights=rep(Nh/n.optlog,n.optlog),
    fpc=rep(n.optlog/Nh,n.optlog),
    data=stsi.data.optlog)
  est.optlog[i] <- svytotal(~LOGVAC,ech.stsi.optlog)[1]
}

```

```
mean(est.optlog)
```

```
## [1] 10753.62
```

```
(cv.optlog <- sd(est.optlog)/mean(est.optlog))
```

```
## [1] 0.07129638
```

Stratification géographique

```

set.seed(123456)
n.sim <- 10000
est.prop <- numeric(n.sim)
for (i in 1:n.sim)
{
  stsi.prop <- sampling::strata(rec99.order,"STRATLOG",
    size=n.prop,method="srswor")
  stsi.data.prop <- getdata(rec99.order,stsi.prop)
  ech.stsi.prop <- svydesign(id=~CODE_N,strata=~STRATLOG,
    weights=rep(Nh/n.prop,n.prop),
    fpc=rep(n.prop/Nh,n.prop),
    data=stsi.data.prop)
  est.prop[i] <- svytotal(~LOGVAC,subset(ech.stsi.prop,NAME_3=="Muret"))[1]
}

```

```
mean(est.prop)
```

```
## [1] 3078.013
```

```
(cv.prop <- sd(est.prop)/mean(est.prop))
```

```
## [1] 0.3694285
```

```
# 0.33
```